

ME1411 Engineering Material Science

Lecture 1 Overview

What is Material Science

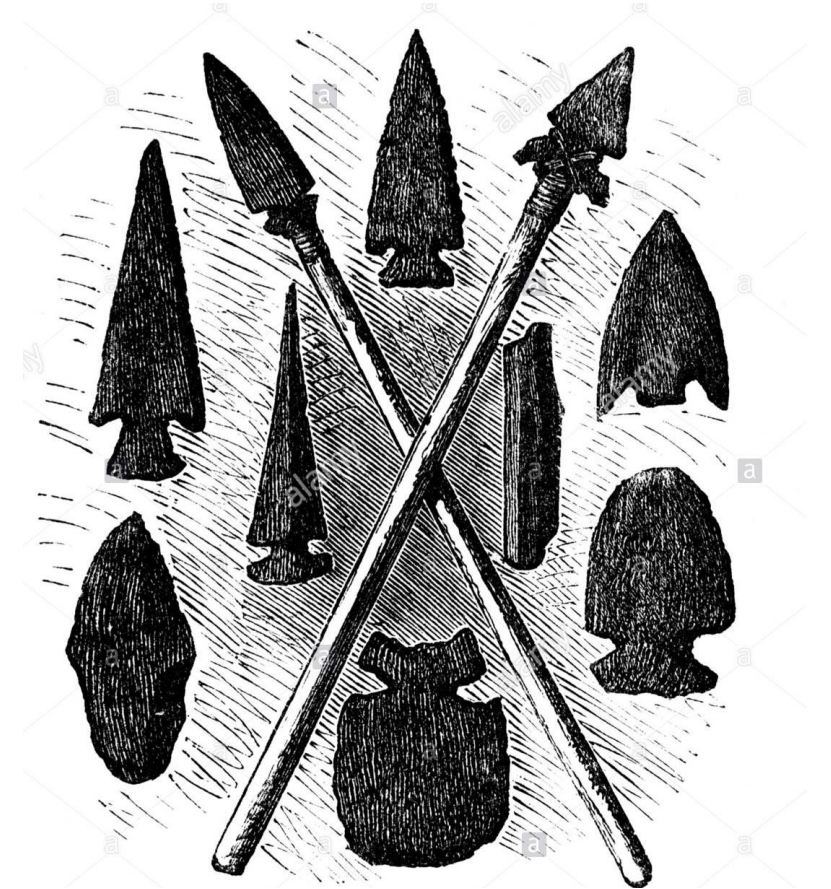


https://www.youtube.com/watch?v=_cUEjPtVIIM

Early Civilization

The development and advancement of human societies have been closely tied to the people's ability to produce and manipulate materials to fill their needs.

Stone Age (2.5 Billion B.C.)

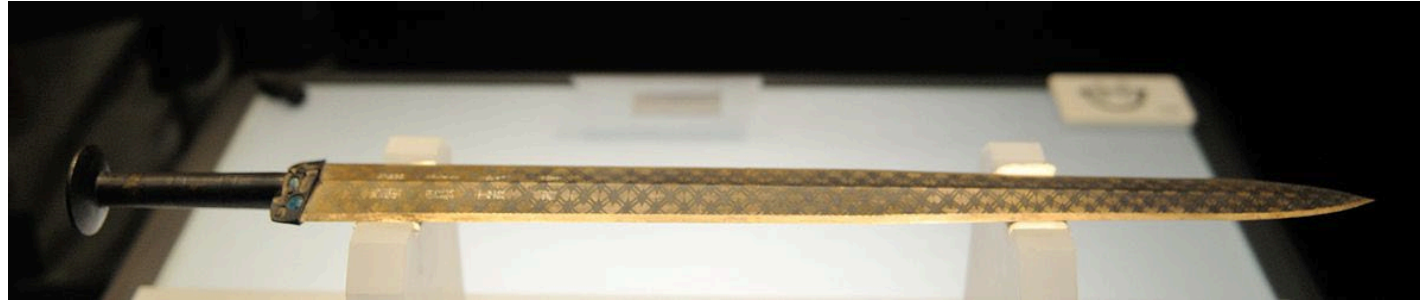


Bronze Age (3500 B.C.)



Bronze is an alloy of copper and tin

Sharp Knife Cuts Hair only by blowing: Sword of a King



- The blade length of less than 1 m; 55.7 cm long, 4.6 cm wide, 8.4 cm long handle.
- Blade overall treatment after curing, the degree of corrosion resistance and toughness has been greatly improved,
- The smelting and metal surface treatment technology has reached a very high level.

<https://www.youtube.com/watch?v=AVVygEH-XXU>

Iron Age (1000 B.C.)



Now?

- Silicon Age?
 - Polymer Age?
 - Nano Age?
-
- Materials— driven force of our society



CPU made of - Silicon

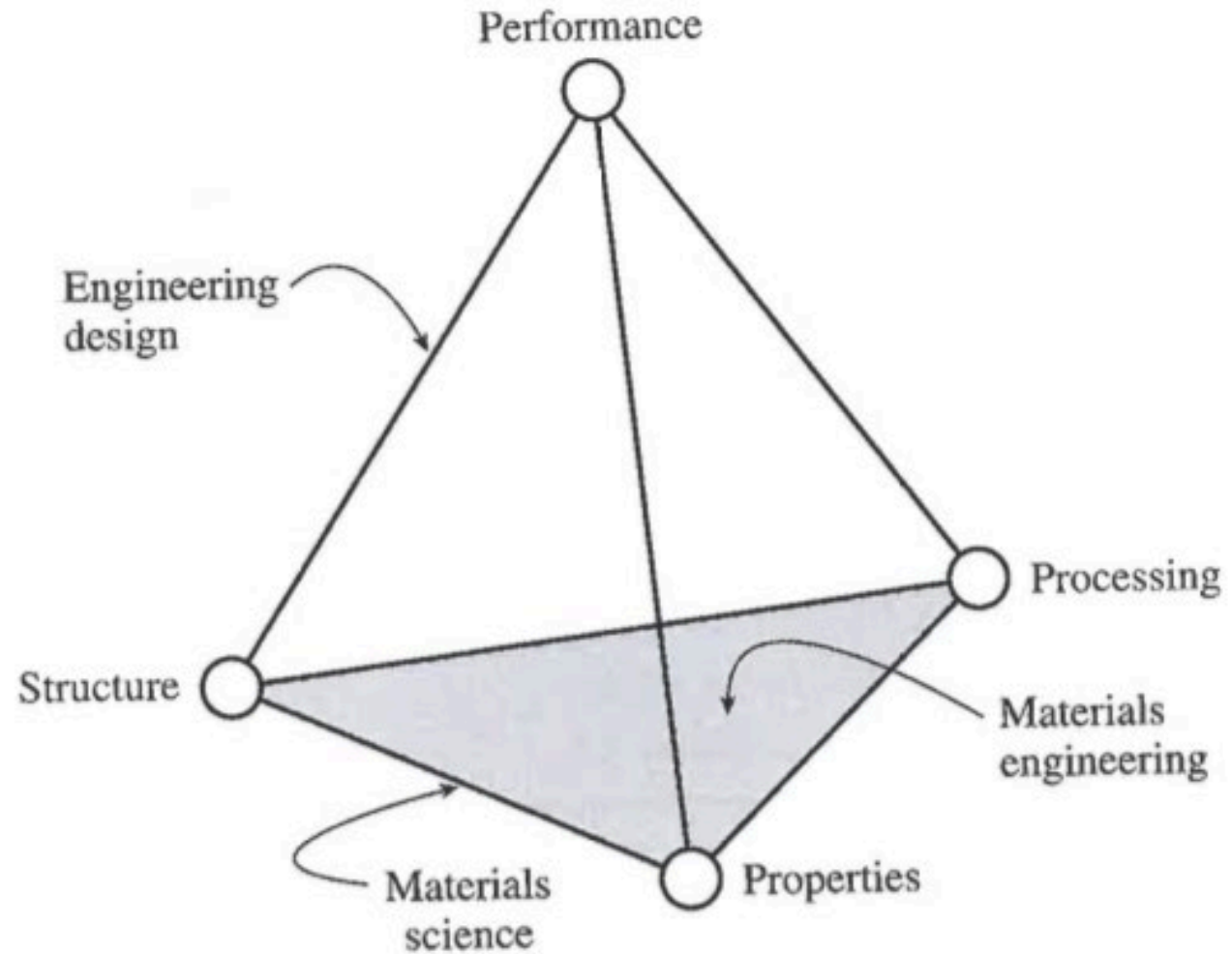
National Museum of American
History, Washington D.C.



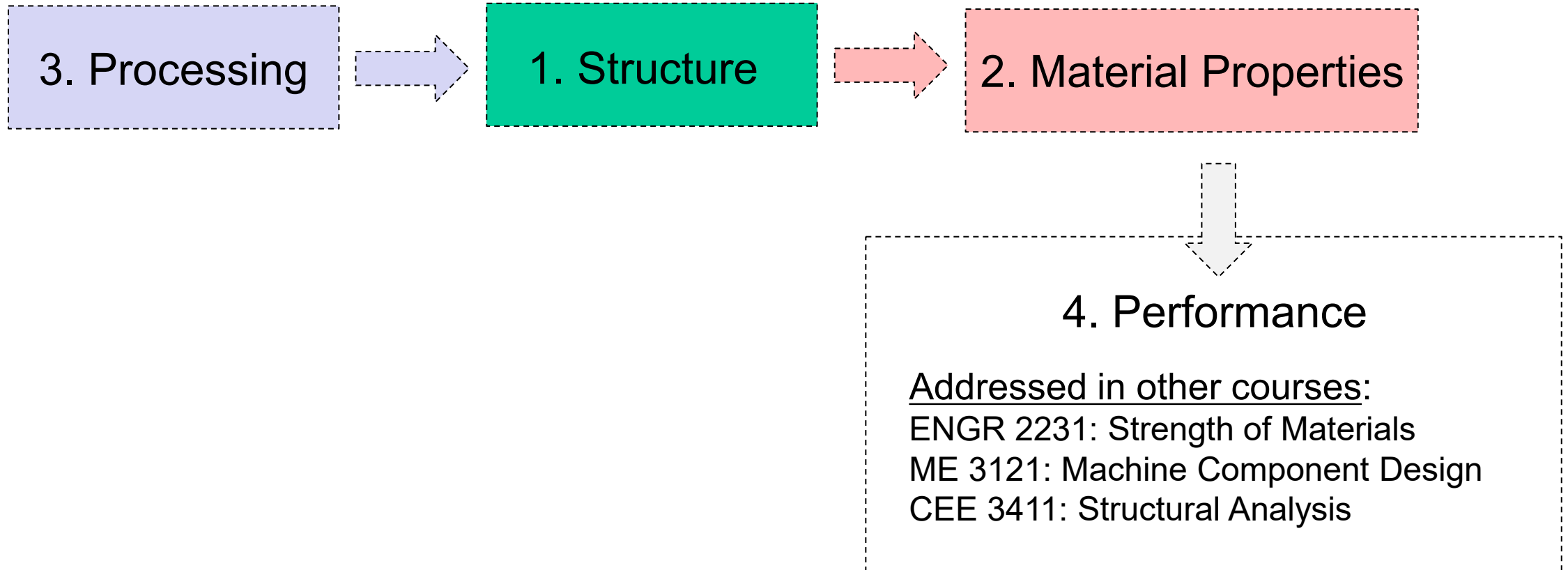
Why Material Science?

- Select the right material for your design
 - No ideal combination of properties
 - Strong-brittle
 - Easy to manufactured or constructed
 - Cost
 - Deterioration

Material Science and Engineering



How will we organize the topics for this course?



Classifications of Materials

Classical Categories:

1. Metals
2. Ceramics
3. Polymers
4. Composites

Classification of Materials

Metals

- composed of one or more metallic elements
- many combined with nonmetallic elements
- stiff, strong, and ductile, high thermal and electrical conductivity, opaque reflective

Classification of Materials

Ceramics

- compounds of metallic and nonmetallic elements
- brittle, glassy, elastic, insulators (non-conducting)
 - Aluminum Oxide - Al_2O_3
 - Silicon Carbide - SiC
 - Iron oxides-
 - Ferrous Oxide (Wursterite): FeO ,
 - Ferric Oxide (Hematite): Fe_2O_3
 - Magnetite: Fe_3O_4

Classification of Materials

Polymers

- very long molecular structures
- soft, ductile, low strength, low density, thermal and electrical insulators
- plastic and rubber materials
- Examples:
 - PC – polycarbonate
 - PVC – polyvinyl chloride

Classification of Materials

Composites

- composed of two or more individual materials
- Examples: concrete, bamboo, wood, bone tissue

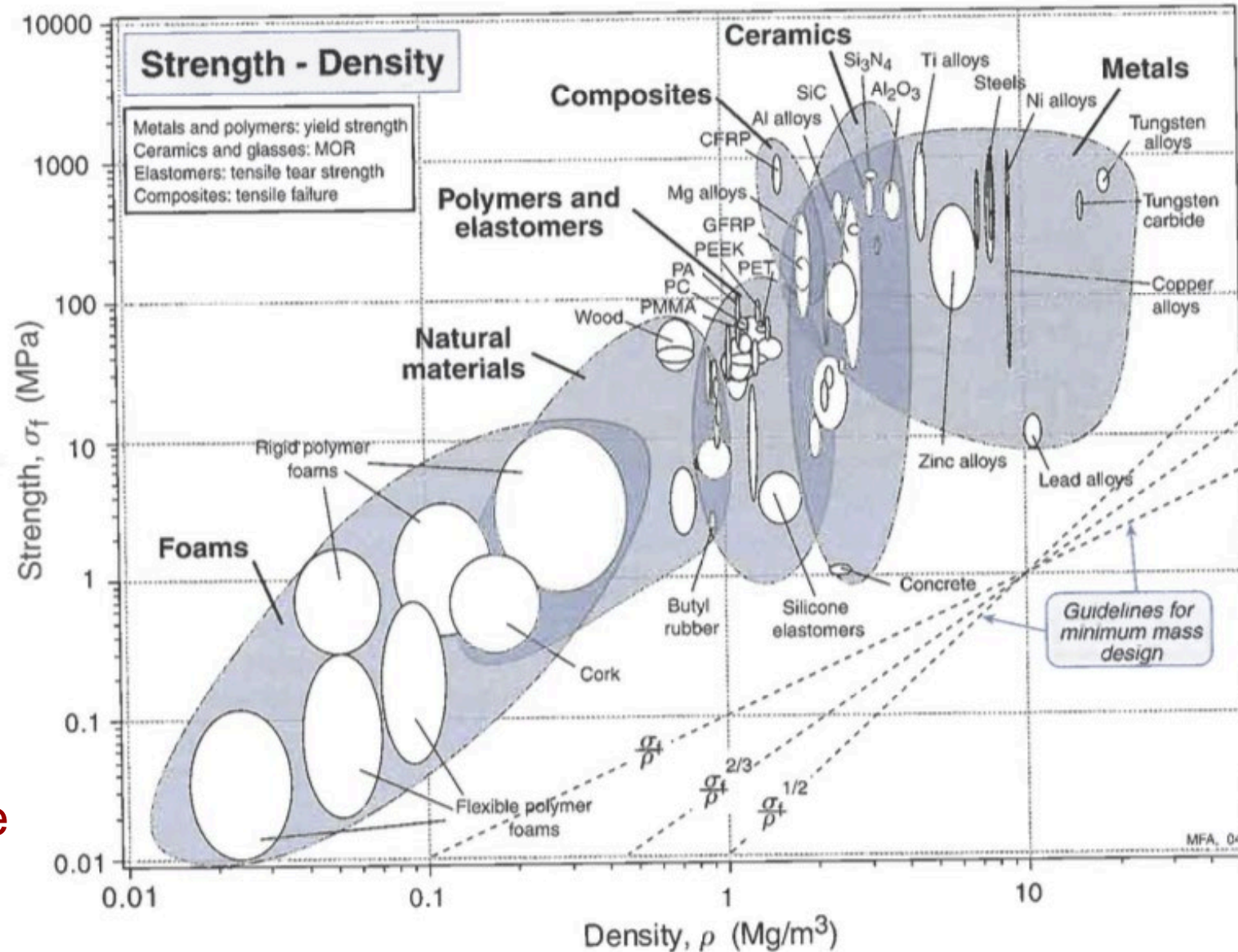
Classification of Materials

Advanced Materials

High tech. application with enhanced properties, might be belong to one of the 4 types of materials or might be all material types

- semiconductor
- biomaterials
- smart materials
- nanoengineered materials

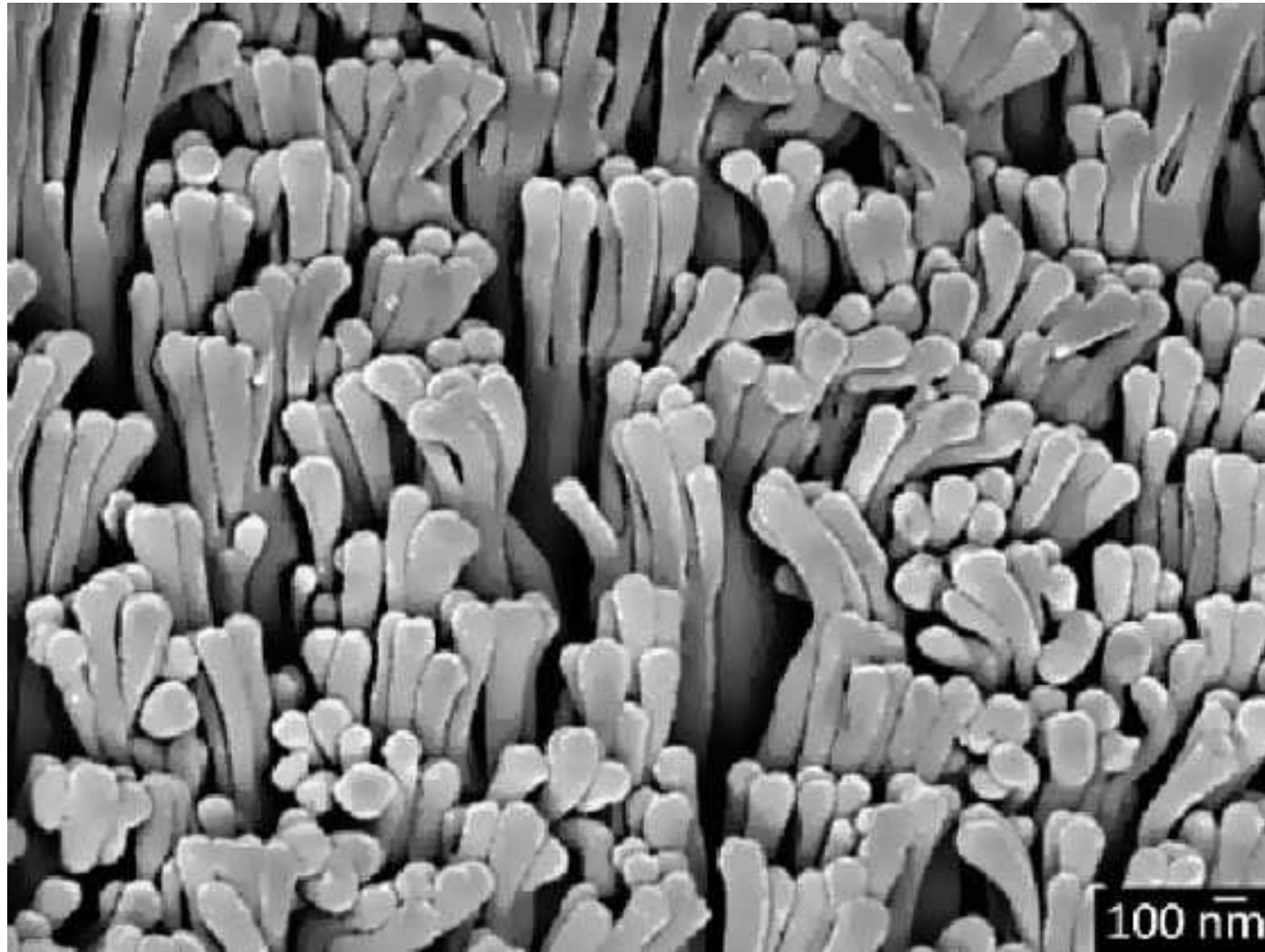
Trend: Strength increases with density



Notice: Axes are in log scale

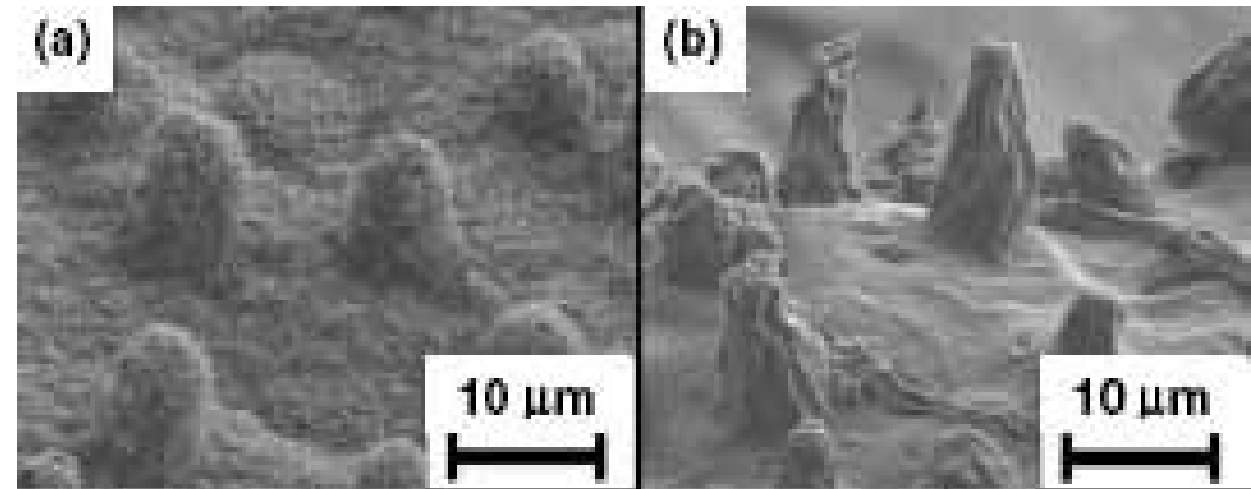


Geckos can rapidly run up vertical walls and along the undersides of horizontal surfaces.



Their secret lies in their toe pads---extremely large number of microscopic small hairs in contact with surface to establish weak forces of attraction.

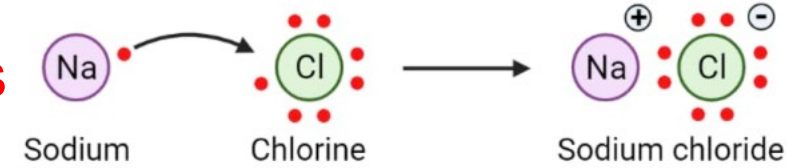
Hydrophobic surface



Chemical Bonding

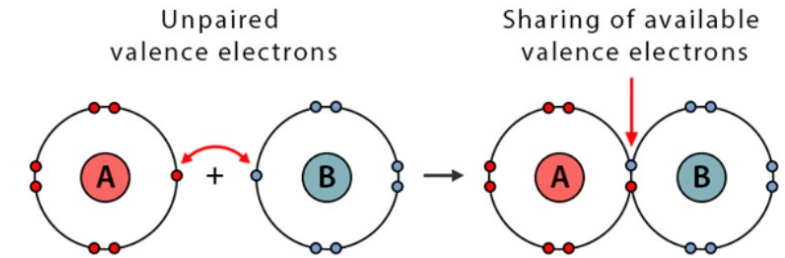
- Ionic

- nondirectional; valence electrons transfer; **ceramics**



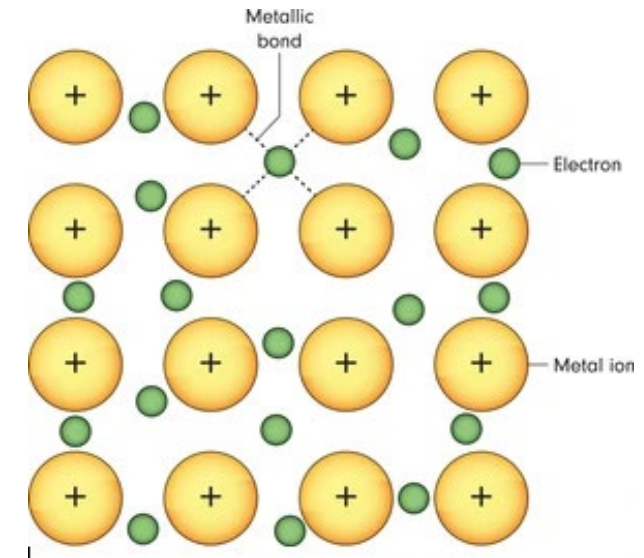
- Covalent

- directional; share electrons
- **ceramics; polymers (within molecular chains)**



- Metallic

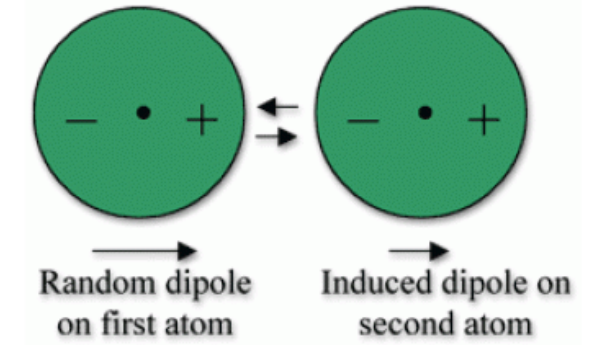
- nondirectional
- an array of positive ions surrounded by a sea of mobile valence electrons
- **metals**



Secondary Bonding

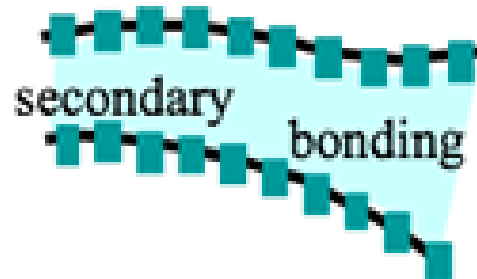
- Hydrogen bonding

- a hydrogen atom located between a pair of other atoms having a high affinity for electrons

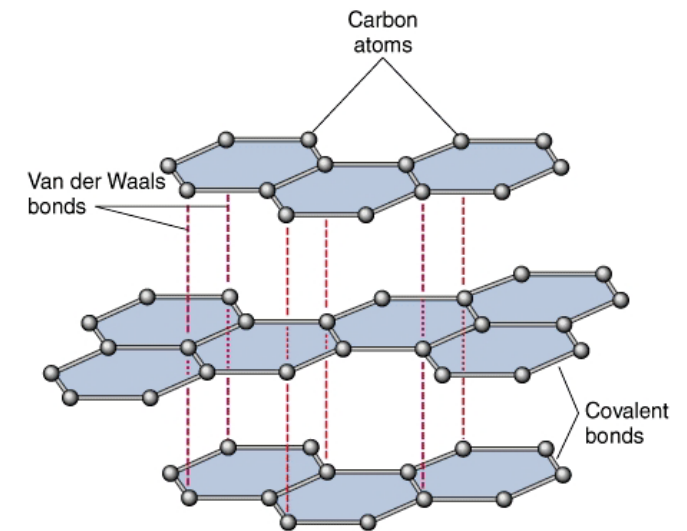


- Van der Waals bonding

- dipole - an uneven charge distribution occurs
- polymers (between molecular chains)
- directional

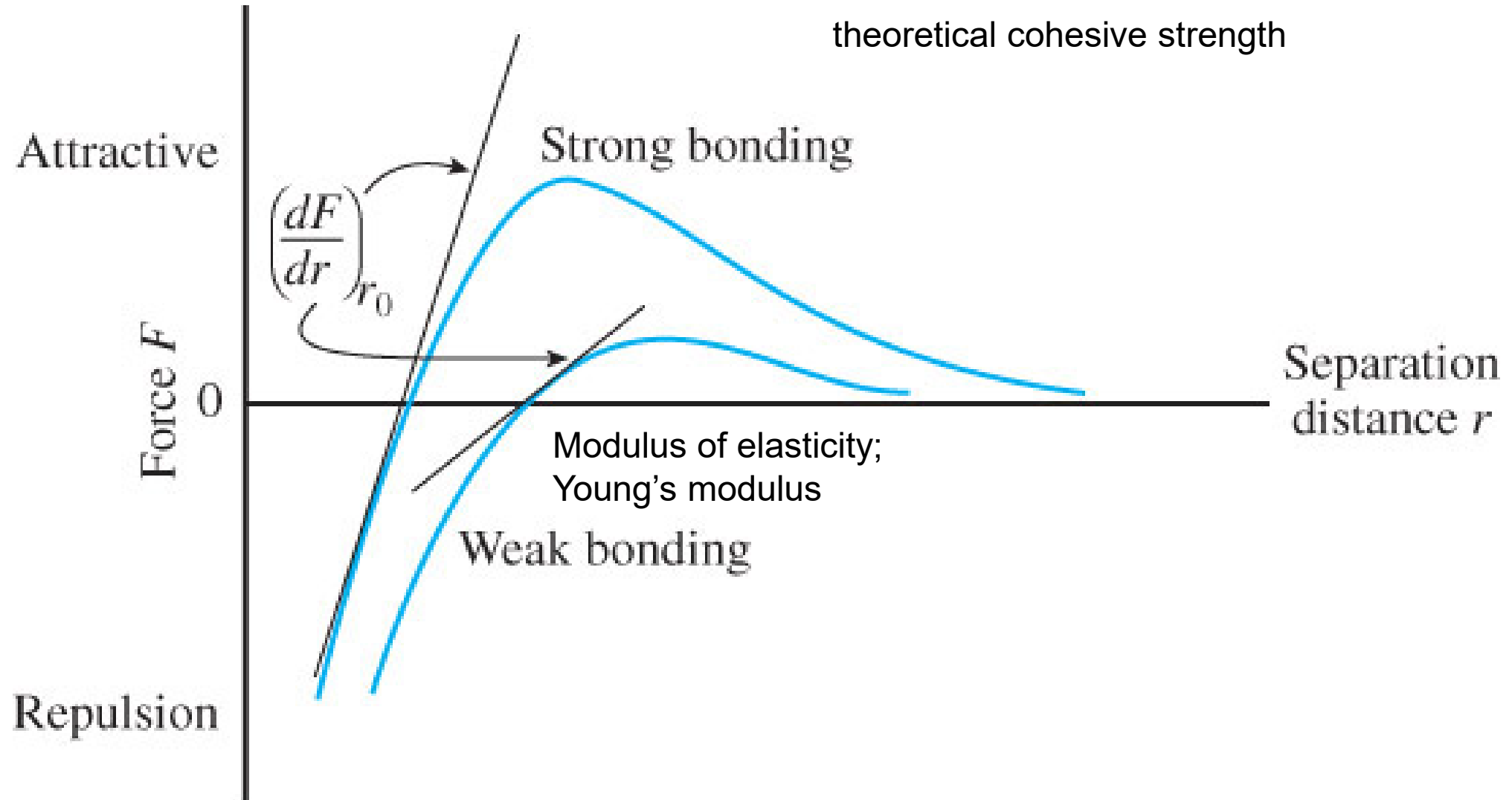


polymer



graphite

Net Bonding Force vs. Separation Distance



Lab Modules

- Pre-Module Assignments –
 - Module 1: Upload to Canvas by the start of class Thursday.
 - Module 2: Upload to Canvas by the start of class Tuesday.
- Lab teammates – Email Dr. Gargac by 11:59 PM today.
- Module 1 → Will be completed during class time
- Module 2 → Your team will sign up for a time slot and will be completed outside of class time. **I will email out instructions with the team assignments.**